FRA – PROJECT PART A & PART B

Finance and Risk Analytics

[Draw your reader in with an engaging abstract. It is typically a short summary of the document. When you're ready to add your content, just click here and start typing.]

Table of Contents

- Part - A
- Executive Summary
- Problem Statement
- Objective
- Data Description
- Sample of the dataset
- Exploratory Data Analysis
- Data Preparation – Split
- Model Building - Logistic Regression
- Model Building - Random Forest
- Model Performance Improvement
- VIF
- ROC Curve
- Part – B
- Overview
- Objective
- EDA
- Statistical Summary

List Of Figures

1. Dataset Sample
2. Statistical Summary of the Dataset
3. Correlation Heatmap
4. Outlier count across data
5. Count of Defaulters
6. Boxplot set 1
7. Boxplot set 2
8. Boxplot set 3
9. Boxplot set 4
10. Boxplot set 5
11. Boxplot set 6
12. Boxplot set 7
13. Boxplot set 8
14. Boxplot set 9
15. Boxplot set 10
16. Test Data
17. Train Data
18. Scaled Data
19. Confusion Matrix | Performance Test data
20. Confusion Matrix | Performance Test data
21. Confusion Matrix
22. Test performance results
23. Confusion Matrix – Random Forest Model - Test Performance
24. Model performance stats
25. Variance Inflation Factor
26. ROC Analysis
27. Confusion Matrix | Model performance stats
28. Confusion Matrix Test set | Model performance stats

Executive Summary

In the realm of modern finance, businesses encounter the perpetual challenge of managing debt obligations effectively to maintain a favourable credit standing and foster sustainable growth. Investors keenly scrutinize companies capable of navigating financial complexities while ensuring stability and profitability. A pivotal instrument in this evaluation process is the balance sheet, which provides a comprehensive overview of a company's assets, liabilities, and shareholder equity, offering insights into its financial health and operational efficiency. In this context, leveraging available financial data, particularly from preceding fiscal periods, becomes imperative for informed decision-making and strategic planning.

Problem Statement

A renowned credit rating organization wants to develop a Financial Health Assessment Tool. With the help of the tool, it endeavours to empower businesses and investors with a robust mechanism for evaluating the financial well-being and creditworthiness of companies. By harnessing machine learning techniques, the organization aims to analyze historical financial statements and extract pertinent insights to facilitate informed decision-making via the tool. Specifically, the organization foresees facilitating the following with the help of the tool:

1. Debt Management Analysis: Identify patterns and trends in debt management practices to assess the ability of businesses to fulfill financial obligations promptly and efficiently and identify potential cases of default.

2. Credit Risk Evaluation: Evaluate credit risk exposure by analyzing liquidity ratios, debt-to-equity ratios, and other key financial indicators to ascertain the likelihood of default and inform investment decisions.

As a part of the data science team in the organization, you have been provided with the financial metrics of different companies. The task is to analyze the data provided and develop a predictive model leveraging machine learning techniques to identify whether a given company will default on its debt repayments in the next two quarters. The predictive model will help the organization anticipate potential challenges with the financial performance of the companies and enable proactive risk mitigation strategies.

Introduction

Objective

The objective of this analysis is to conduct Market Risk Analysis on a portfolio of Indian stocks using Python. It uses historical stock price data to understand market volatility and riskiness. Using statistical measures like mean and standard deviation, investors gain a deeper understanding of individual stocks' performance and portfolio variability.

Through this analysis, investors can aim to achieve the following objectives:

- Risk Assessment: Analyze the historical volatility of individual stocks and the overall portfolio.
- Portfolio Optimization: Use Market Risk Analysis insights to enhance risk-adjusted returns.
- Performance Evaluation: Assess portfolio management strategies' effectiveness in mitigating market risk.
- Portfolio Performance Monitoring: Monitor portfolio performance over time and adjust as market conditions and risk preferences change.

Data Description

The dataset contains weekly stock price data for 5 Indian stocks over an 8-year period. The dataset enables us to analyze the historical performance of individual stocks and the overall market dynamics.

Co_Code	Company Code
Co_Name	Company Name
_Operating_Expense_Rate	Operating Expense Rate: Operating Expenses/Net Sales. The operating expense ratio (OER) is the cost to operate a piece of property compared to the income the property brings in.
_Research_and_development_expense_rate	Research and development expense rate: (Research and Development Expenses)/Net Sales. Research and development (R&D) expenses are direct expenditures relating to a company's efforts to develop, design, and enhance its products, services, technologies, or processes.
_Cash_flow_rate	Cash flow rate: Cash Flow from Operating/Current Liabilities. Cash flow is a measure of how much cash a business brought in or spent in total over a period.
_Interest_bearing_debt_interest_rate	Interest-bearing debt interest rate: Interest-bearing Debt/Equity
_Tax_rate_A	Tax rate (A): Effective Tax Rate. Effective tax rate represents the percentage of their taxable income that individuals pay in taxes. For corporations, the effective corporate tax rate is the rate they pay on their pre-tax profits.
_Cash_Flow_Per_Share	Cash Flow Per Share. It is the after-tax earnings plus depreciation on a per-share basis that functions as a measure of a firm's financial strength
_Per_Share_Net_profit_before_tax_Yuan_	Per Share Net profit before tax (Yuan ¥): Pretax Income Per Share. Pretax income, also known as earnings before tax or pretax earnings, is the net income earned by a business before taxes are subtracted/accounted for.
_Realized_Sales_Gross_Profit_Growth_Rate	Realized Sales Gross Profit Growth Rate.
_Operating_Profit_Growth_Rate	Operating Profit Growth Rate: Operating Income Growth. It is the rate of increase in operating income over the last year.
_Continuous_Net_Profit_Growth_Rate	Continuous Net Profit Growth Rate: Net Income-Excluding Disposal Gain or Loss Growth
_Total_Asset_Growth_Rate	Total Asset Growth Rate: Total Asset Growth. It is the rate at which how quickly the company has been growing its Assets
_Net_Value_Growth_Rate	Net Value Growth Rate: Total Equity Growth
_Total_Asset_Return_Growth_Rate_Ratio	Total Asset Return Growth Rate Ratio: Return on Total Asset Growth
_Cash_Reinvestment_perc	Cash Reinvestment %: Cash Reinvestment Ratio. It is the valuation ratio that is used to measure the percentage of annual cash flow that the company invests back into the business as a new investment.

_Current_Ratio	Current Ratio. The current ratio describes the relationship between a company's assets and liabilities
_Quick_Ratio	Quick Ratio: Acid Test. Acid-test ratio (also known as quick ratio) is a measure of a company's liquidity, which is its ability to pay its short-term obligations using only its most liquid assets.
_Interest_Expense_Ratio	Interest Expense Ratio: Interest Expenses/Total Revenue
_Total_debt_to_Total_net_worth	Total debt/Total net worth: Total Liability/Equity Ratio
_Long_term_fund_suitability_ratio_A	Long-term fund suitability ratio (A): (Long-term Liability+Equity)/Fixed Assets
_Net_profit_before_tax_to_Paid_in_capital	Net profit before tax/Paid-in capital: Pretax Income/Capital
_Total_Asset_Turnover	Total Asset Turnover. Net Sales/Average Total Assets
_Accounts_Receivable_Turnover	Accounts Receivable Turnover. The accounts receivable turnover ratio, or receivables turnover, is used in business accounting to quantify how well companies are managing the credit that they extend to their customers by evaluating how long it takes to collect the outstanding debt throughout the accounting period.

Sample of the dataset

	Co_Code	Co_Name	_Operating_Expense_Rate	_Research_and_development_expense_rate	_Cash_flow_rate	_Interest_bearing_debt_interest_rate	_Tax_rate_A	_Cash_Flow_Per_Share	_Per_Shar
0	16974	Hind.Cables	8.820000e+09	0.000000e+00	0.462045	0.000352	0.001417	0.322558	
1	21214	Tata Tele. Mah.	9.380000e+09	4.230000e+09	0.460116	0.000716	0.000000	0.315520	
2	14852	ABG Shipyard	3.800000e+09	8.150000e+08	0.449893	0.000496	0.000000	0.299851	
3	2439	GTL	6.440000e+09	0.000000e+00	0.462731	0.000592	0.009313	0.319834	
4	23505	Bharati Defence	3.680000e+09	0.000000e+00	0.463117	0.000782	0.400243	0.325104	

5 rows × 58 columns

Fig – 1 Dataset Sample

The dataset contains weekly stock price data for 5 Indian stocks over an 8-year period. The dataset enables us to analyze the historical performance of individual stocks and the overall market dynamics.

Exploratory Data Analysis

- The dataset contains 2058 entries (non-null), each representing a different company.
- The dataset has 58 columns out of which 5 are *int64* and rest are *float64*.
- Each company is identified by a unique Co_Code and Co_Name.
- The dataset includes various financial ratios and rates such as Operating_Expense_Rate, Research_and_development_expense_rate, Cash_flow_rate, Interest_bearing_debt_interest_rate, Tax_rate_A, and many others. These metrics provide insights into the company's operational efficiency, financial stability, and profitability.
- There are also metrics related to the company's assets, liabilities, and equity such as Total_Asset_Growth_Rate, Net_Value_Growth_Rate, Total_debt_to_Total_net_worth, Equity_to_Liability, and others. These can help in understanding the company's financial structure and growth potential.
- The dataset also includes some flags like Liability_Assets_Flag, Net_Income_Flag, and Default which seem to indicate certain financial conditions or events for the companies.

Statistical summary Sample

	count	mean	std	min	25%	50%	75%	max
Co_Code	2058.0	1.757211e+04	2.189289e+04	4.000000	3.674000e+03	6.240000e+03	2.428075e+04	7.249300e+04
Operating_Expense_Rate	2058.0	2.052389e+09	3.252624e+09	0.000100	1.578727e-04	3.330330e-04	4.110000e+09	9.980000e+09
Research_and_development_expense_rate	2058.0	1.208634e+09	2.144568e+09	0.000000	0.000000e+00	1.994130e-04	1.550000e+09	9.980000e+09
Cash_flow_rate	2058.0	4.652426e-01	2.266269e-02	0.000000	4.600991e-01	4.634450e-01	4.680691e-01	1.000000e+00
Interest_bearing_debt_interest_rate	2058.0	1.113022e+07	9.042595e+07	0.000000	2.760280e-04	4.540450e-04	6.630660e-04	9.900000e+08
Tax_rate_A	2058.0	1.147770e-01	1.524457e-01	0.000000	0.000000e+00	3.709890e-02	2.161909e-01	9.996963e-01
Cash_Flow_Per_Share	1891.0	3.199856e-01	1.529979e-02	0.169449	3.149890e-01	3.206479e-01	3.259178e-01	4.622268e-01
Per_Share_Net_profit_before_tax_Yuan_	2058.0	1.769673e-01	3.015730e-02	0.000000	1.666039e-01	1.756421e-01	1.858854e-01	7.923477e-01
Realized_Sales_Gross_Profit_Growth_Rate	2058.0	2.276117e-02	2.170104e-02	0.004282	2.205831e-02	2.210001e-02	2.215200e-02	1.000000e+00
Operating_Profit_Growth_Rate	2058.0	8.481083e-01	4.589093e-03	0.736430	8.479740e-01	8.480386e-01	8.481147e-01	1.000000e+00
Continuous_Net_Profit_Growth_Rate	2058.0	2.173915e-01	5.678779e-03	0.000000	2.175741e-01	2.175961e-01	2.176198e-01	2.332046e-01
Total_Asset_Growth_Rate	2058.0	5.287663e+09	2.912615e+09	0.000000	4.315000e+09	6.225000e+09	7.220000e+09	9.980000e+09
Net_Value_Growth_Rate	2058.0	5.189504e+06	2.077918e+08	0.000000	4.362833e-04	4.554170e-04	4.883758e-04	9.330000e+09
Total_Asset_Return_Growth_Rate_Ratio	2058.0	2.641004e-01	2.415661e-03	0.251620	2.637383e-01	2.640161e-01	2.643097e-01	3.586288e-01
Cash_Reinvestment_perc	2058.0	3.771970e-01	2.737311e-02	0.025828	3.707295e-01	3.789678e-01	3.855575e-01	1.000000e+00
Current_Ratio	2058.0	1.336249e+06	6.061917e+07	0.000000	6.567062e-03	8.945370e-03	1.350542e-02	2.750000e+09
Quick_Ratio	2058.0	2.775510e+07	4.448654e+08	0.000000	2.946399e-03	5.284241e-03	8.902983e-03	9.230000e+09
Interest_Expense_Ratio	2058.0	6.312913e-01	6.785512e-03	0.525126	6.306116e-01	6.307999e-01	6.317437e-01	8.121652e-01
Total_debt_to_Total_net_worth	2037.0	1.071429e+07	2.696960e+08	0.000000	3.924894e-03	7.270721e-03	1.306869e-02	9.940000e+09
Long_term_fund_suitability_ratio_A	2058.0	8.973310e-03	3.485186e-02	0.004129	5.162031e-03	5.517000e-03	6.415301e-03	1.000000e+00
Net_profit_before_tax_to_Paid_in_capital	2058.0	1.753994e-01	2.622348e-02	0.000000	1.658623e-01	1.745683e-01	1.844450e-01	7.921047e-01
Total_Asset_Turnover	2058.0	1.286405e-01	1.006216e-01	0.000000	6.146927e-02	1.034483e-01	1.679160e-01	9.190405e-01
Accounts_Receivable_Turnover	2058.0	4.159864e+07	5.047673e+08	0.000000	7.446260e-04	1.081432e-03	1.854463e-03	9.740000e+09
Average_Collection_Days	2058.0	2.629786e+07	4.109967e+08	0.000000	3.576384e-03	6.001272e-03	8.638997e-03	8.800000e+09
Inventory_Turnover_Rate_times	2058.0	2.030227e+09	3.077250e+09	0.000000	1.909297e-04	1.910000e+07	3.815000e+09	9.990000e+09

Fig – 2 Statistical Summary of the Dataset

Correlation Heatmap

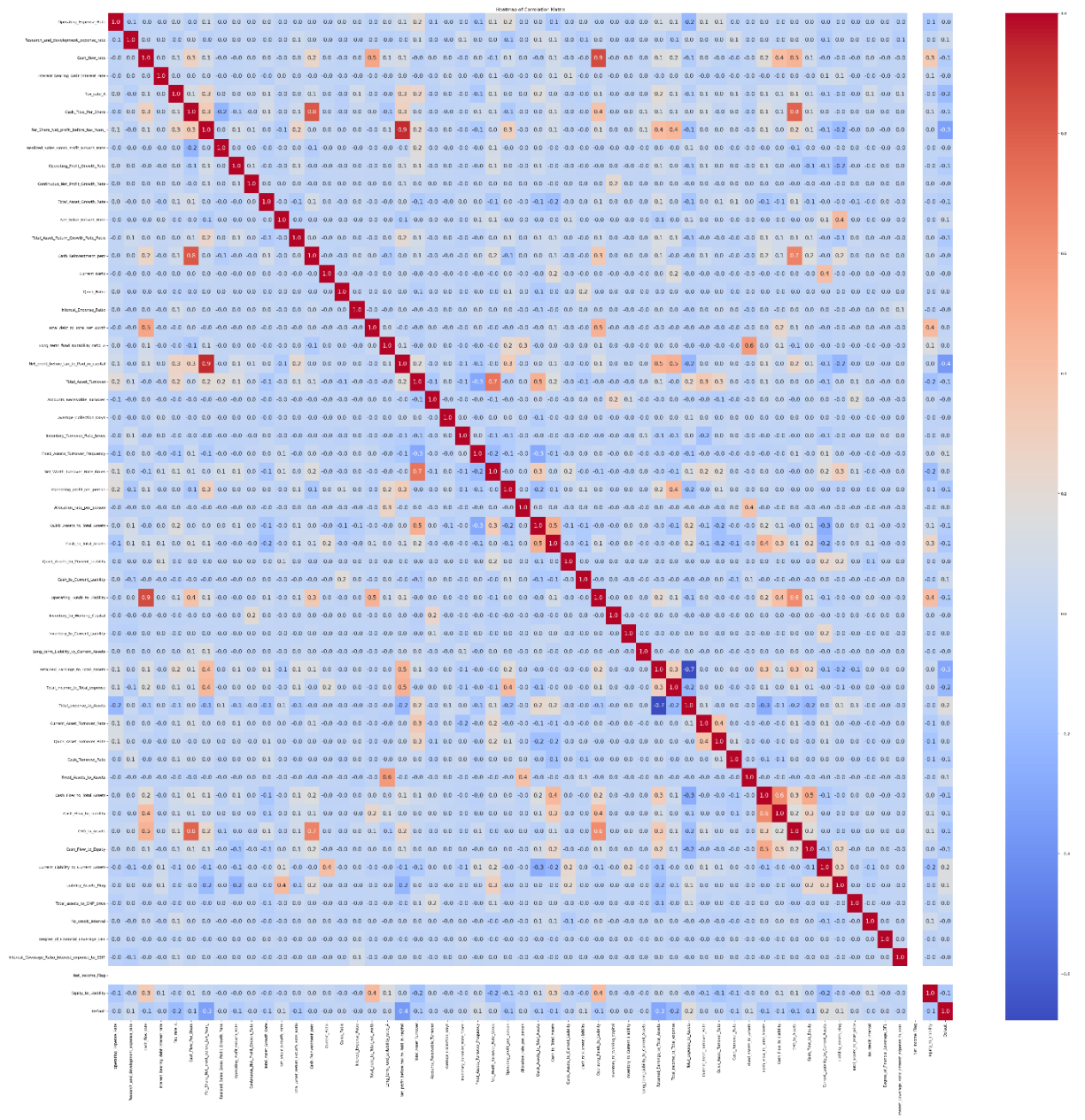


Fig – 3 Correlation Heatmap

In this heatmap, each cell represents the correlation between two variables, with the variable names indicated on the x and y axes. The color of each cell represents the strength and direction of the correlation:

- Dark red cells represent a strong positive correlation, meaning that as one variable increases, the other also tends to increase.
- Lighter cells represent weaker correlations.

Outlier Detection & Count

	Column	No. of outliers
0	Operating_Expense_Rate	0
1	Research_and_development_expense_rate	264
2	Cash_flow_rate	206
3	Interest_bearing_debt_interest_rate	94
4	Tax_rate_A	42
5	Cash_Flow_Per_Share	146
6	Per_Share_Net_profit_before_tax_Yuan_	186
7	Realized_Sales_Gross_Profit_Growth_Rate	283
8	Operating_Profit_Growth_Rate	317
9	Continuous_Net_Profit_Growth_Rate	340
10	Total_Asset_Growth_Rate	0
11	Net_Value_Growth_Rate	304
12	Total_Asset_Return_Growth_Rate_Ratio	226
13	Cash_Reinvestment_perc	220
14	Current_Ratio	193
15	Quick_Ratio	190
16	Interest_Expense_Ratio	328
17	Total_debt_to_Total_net_worth	105
18	Long_term_fund_suitability_ratio_A	234
19	Net_profit_before_tax_to_Paid_in_capital	173
20	Total_Asset_Turnover	101
21	Accounts_Receivable_Turnover	281
22	Average_Collection_Days	77
23	Inventory_Turnover_Rate_times	29
24	Fixed_Assets_Turnover_Frequency	501
25	Net_Worth_Turnover_Rate_times	165
26	Operating_profit_per_person	357
27	Allocation_rate_per_person	200
28	Quick_Assets_to_Total_Assets	4
29	Cash_to_Total_Assets	163
30	Quick_Assets_to_Current_Liability	185
31	Cash_to_Current_Liability	253
32	Operating_Funds_to_Liability	219
33	Inventory_to_Working_Capital	247
34	Inventory_to_Current_Liability	129
35	Long_term_Liability_to_Current_Assets	213
36	Retained_Earnings_to_Total_Assets	208
37	Total_income_to_Total_expense	136
38	Total_expense_to_Assets	168
39	Current_Asset_Turnover_Rate	464
40	Quick_Asset_Turnover_Rate	0
41	Cash_Turnover_Rate	0
42	Fixed_Assets_to_Assets	10
43	Cash_Flow_to_Total_Assets	317
44	Cash_Flow_to_Liability	407
45	CFO_to_Assets	110
46	Cash_Flow_to_Equity	306
47	Current_Liability_to_Current_Assets	121
48	Total_assets_to_GNP_price	235
49	No_credit_Interval	396
50	Degree_of_Financial_Leverage_DFL	438
51	Interest_Coverage_Ratio_Interest_expense_to_EBIT	376
52	Equity_to_Liability	190
53	Default	220

Fig – 4 Outlier count across data

Here are some insights from the Correlation table:

- The Operating_Expense_Rate, Total_Asset_Growth_Rate, Quick_Asset_Turnover_Rate, and Cash_Turnover_Rate columns have no outliers. This suggests that the values in these columns are relatively evenly distributed.
- The Fixed_Assets_Turnover_Frequency and Current_Asset_Turnover_Rate columns have the highest number of outliers, with 501 and 464 outliers respectively. This indicates that these columns have several values that are significantly different from the rest.
- The Research_and_development_expense_rate and Continuous_Net_Profit_Growth_Rat e columns also have a high number of outliers. This could suggest that there are a few companies with unusually high or low research and development expense rates and net profit growth rates.
- The Inventory_Turnover_Rate_times and Quick_Assets_to_Total_Assets columns have the least number of outliers, with 29 and 4 outliers respectively. This suggests that these columns are relatively evenly distributed, with most values falling within the expected range.

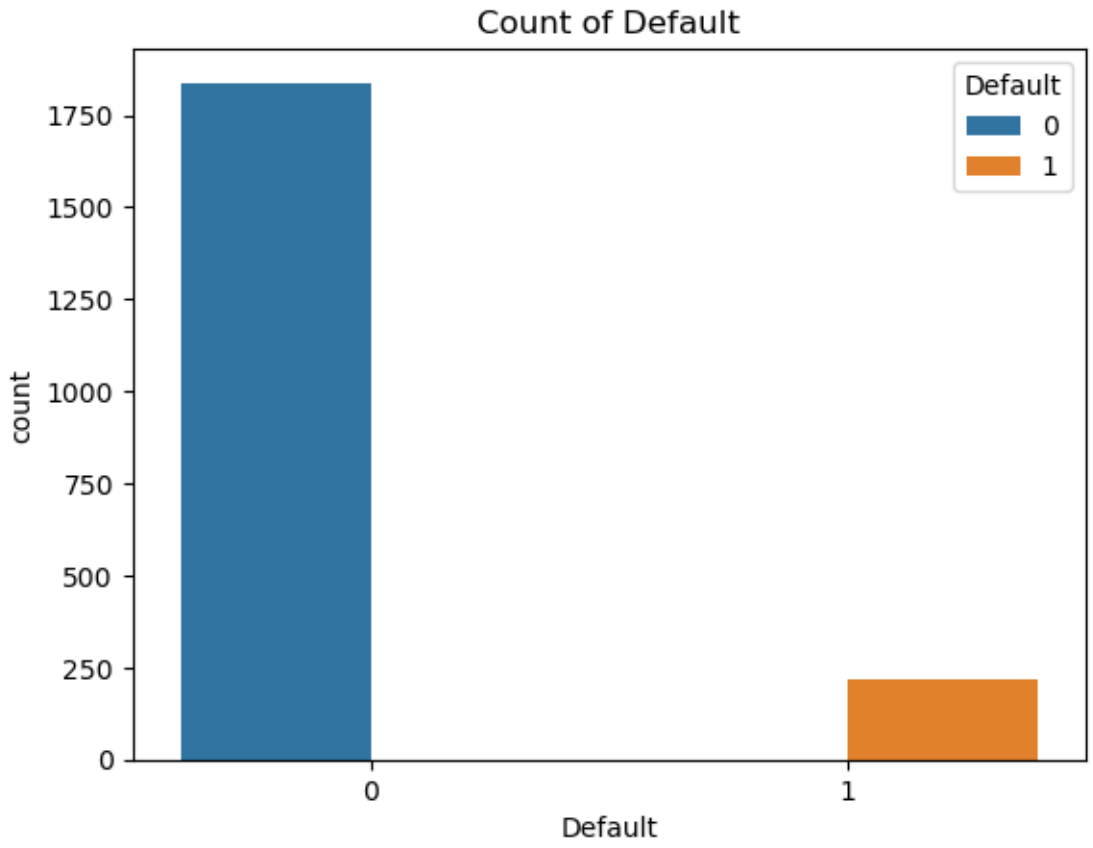


Fig – 5 Count of Defaulters

The countplot is showing the frequency of ‘Defaulters’ and ‘Non-Defaulters’ in your dataset.

Here’s what the countplot tells us:

- The x-axis represents two categories: ‘0’ and ‘1’. In the context of your question, ‘0’ likely represents ‘non-defaulters’ and ‘1’ represents ‘Defaulters’.
- The y-axis represents the count or frequency of each category in your dataset.
- The blue bar represents the ‘non-defaulters’ (category ‘0’), and the orange bar represents the ‘Defaulters’ (category ‘1’).
- The height of the blue bar is around 1750, indicating that there are approximately 1750 ‘non-defaulters’ in your dataset.
- The height of the orange bar is just above 250, indicating that there are slightly more than 250 ‘Defaulters’ in your dataset.
- From these observations, we can infer that ‘non-defaulters’ significantly outnumber ‘Defaulters’ in your dataset.
- Percentage of defaulters as per calculations is 10.689990281827017.

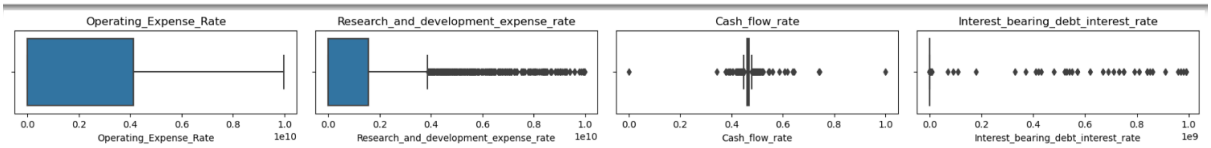


Fig – 6 Boxplot set 1

1. Operating Expense Ratio: This boxplot has a wide interquartile range (IQR), indicating a significant variability within the middle 50% of data points. The median is towards the lower end of the box, suggesting that more data points fall below the median than above it. There are no visible outliers.
2. Research and Development Expense Ratio: This boxplot shows a narrower IQR, indicating less variability in this metric. The median is closer to the center of the box, implying a more symmetric distribution of data. There are several outliers in both directions.
3. Cash Flow Ratio: This boxplot has an even narrower IQR and includes several outliers on both sides as well. The median appears to be slightly above center within this narrow IQR.
4. Interest Bearing Debt Interest Rate: This boxplot has an extremely narrow IQR with many outliers on both sides. The median seems to be centrally located within this very tight range

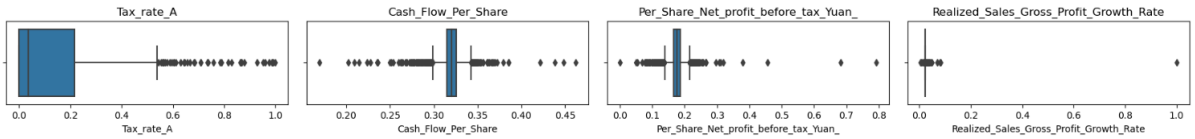


Fig- 7 Boxplot set 2

1. Operating Expense Ratio: This boxplot has a wide interquartile range (IQR), indicating a significant variability within the middle 50% of data points. The median is towards the lower end of the box, suggesting that more data points fall below the median than above it. There are no visible outliers.
2. Research and Development Expense Ratio: This boxplot shows a narrower IQR, indicating less variability in this metric. The median is closer to the center of the box, implying a more symmetric distribution of data. There are several outliers in both directions.
3. Cash Flow Ratio: This boxplot has an even narrower IQR and includes several outliers on both sides as well. The median appears to be slightly above center within this narrow IQR.
4. Interest Bearing Debt Interest Rate: This boxplot has an extremely narrow IQR with many outliers on both sides. The median seems to be centrally located within this very tight range.

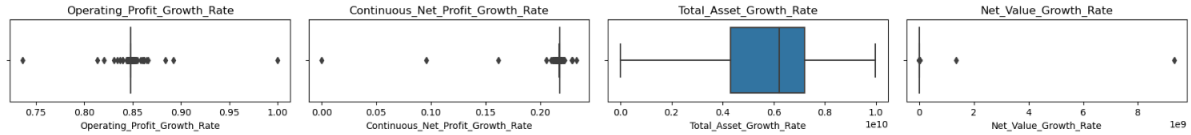


Fig – 8 Boxplot set 3

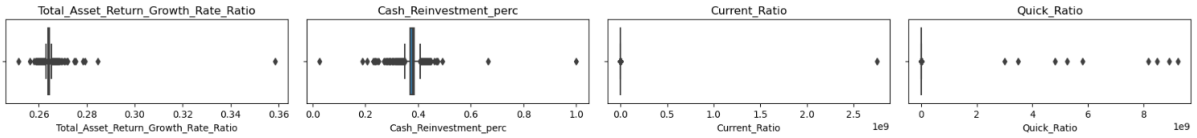


Fig – 9 Boxplot set 4

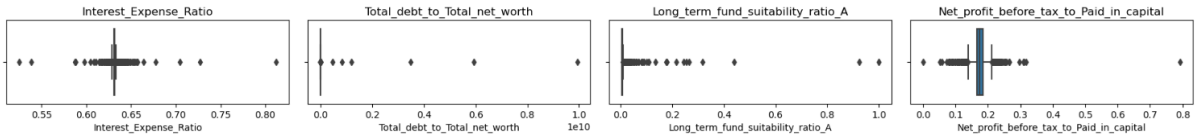


Fig – 10 Boxplot set 5

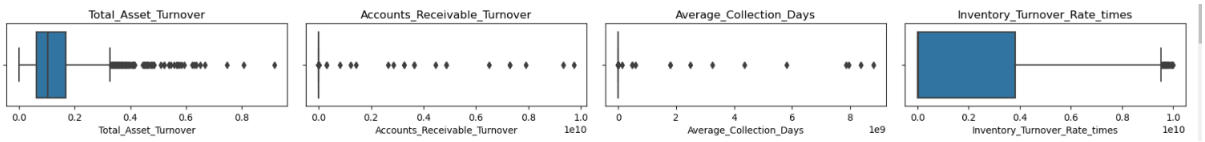


Fig – 11 Boxplot set 6

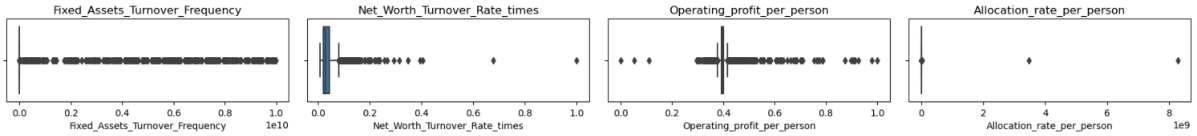


Fig – 12 Boxplot set 7

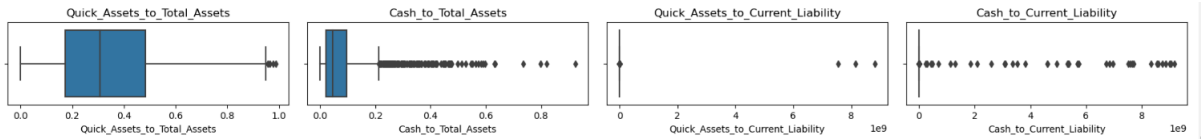


Fig – 13 Boxplot set 8

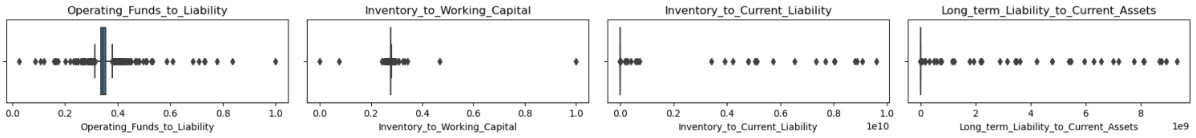


Fig – 14 Boxplot set 9

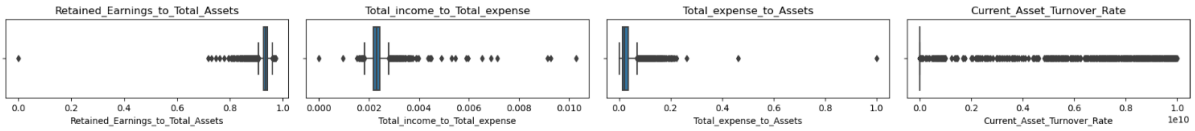


Fig – 15 Boxplot set 10

Retained Earnings to Total Assets:

Insights: The high skewness and outliers suggest a wide disparity in how companies manage their retained earnings. Most companies may be reinvesting less or distributing more of their profits, while a few are retaining a large portion of their earnings.

Total Income to Total Expense:

Insights: The narrow IQR and right skew suggest that while most companies operate with a small profit margin, a few companies manage to generate a higher income relative to their expenses. This could point to efficient cost management or higher profitability in those outliers.

Total Expense to Assets:

- Insights: The relatively low expense-to-asset ratio for most companies suggests efficient asset utilization. The outliers may indicate companies that are either in a high-expense phase (such as scaling operations) or possibly struggling with cost control.

Skewness: All four box plots exhibit right skewness, indicating that most companies are on the lower end of the ratios, with a few companies performing exceptionally well.

Outliers: The presence of numerous outliers in each box plot highlights significant variability in company performance.

Splitting Data for model Testing

Data Split

Test Data

```
1463            0.617529                                0.026287
1398            0.623563                                0.044600
146             0.622404                                0.030641
338             0.624533                                0.026929

      Interest_Coverage_Ratio_Interest_expense_to_EBIT  Equity_to_Liability
62                                            0.565093             0.020433
1144                                          0.561645             0.054822
1612                                          0.565946             0.024906
1149                                          0.565116             0.017808
40                                            0.565302             0.021722
...                                                ...                  ...
240                                           0.554413             0.022168
1463                                          0.560558             0.018304
1398                                          0.569915             0.032626
146                                           0.569112             0.034079
338                                           0.565741             0.044550

[515 rows x 53 columns]
```

Fig – 16 Test Data

Train Data

```
583             0.623906                                0.026792
1058            0.623784                                0.026836
562             0.618200                                0.026773
1400            0.623540                                0.026714

      Interest_Coverage_Ratio_Interest_expense_to_EBIT  Equity_to_Liability
939                                           0.564618             0.016470
540                                           0.565130             0.030487
76                                            0.566128             0.027051
179                                           0.561557             0.013225
226                                           0.564829             0.023284
...                                                ...                  ...
1127                                          0.565479             0.022271
583                                           0.565161             0.065833
1058                                          0.565362             0.054494
562                                           0.565071             0.017965
1400                                          0.564760             0.017277

[1543 rows x 53 columns]
```

Fig - 17 Train Data

Scaled Data

```
X_train_scaled.head()
```

	Operating_Expense_Rate	Research_and_development_expense_rate	Cash_flow_rate	Interest_bearing_debt_interest_rate	Tax_rate_A	Cash_Flow_Per_Share	Per_Share_Net_profit_before_tax_Yuan_	Realized_Sales_Gross_Profit_Growth_Rate	Operating_Pro
0	-0.633296	-0.396806	-0.132455	-0.128462	-0.754347	0.088170	-0.902611	-0.156311	
1	-0.633296	-0.561672	-0.934352	-0.128462	-0.754347	-1.224514	-1.111709	-10.373320	
2	-0.633296	0.361946	-0.290335	-0.128462	0.061964	-0.409659	0.227000	-0.087271	
3	-0.633296	-0.561672	-0.179548	-0.128462	-0.754347	-0.077773	-0.482011	-0.188306	
4	-0.633296	-0.561672	-0.123892	-0.128462	-0.754347	-0.168422	-0.768019	-0.247137	

5 rows × 53 columns

```
X_test_scaled.head()
```

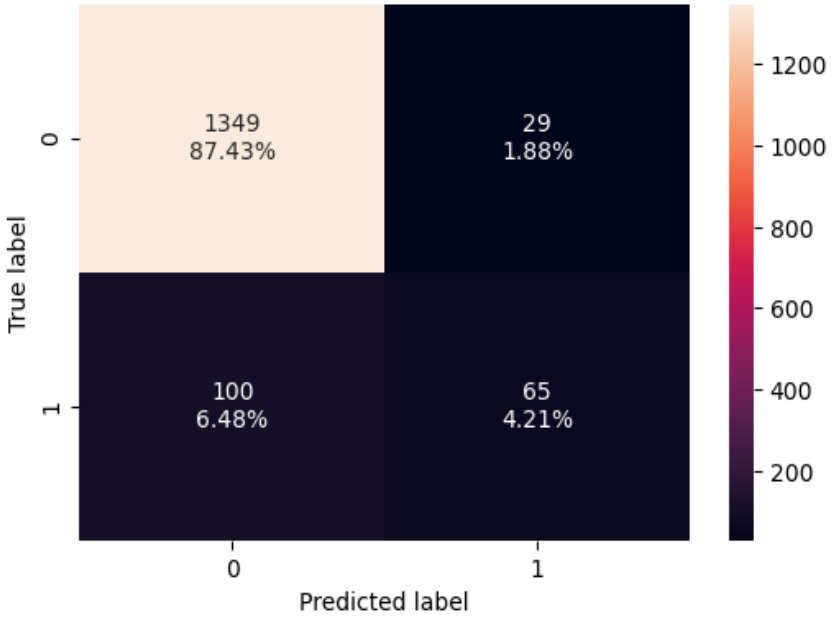
	Operating_Expense_Rate	Research_and_development_expense_rate	Cash_flow_rate	Interest_bearing_debt_interest_rate	Tax_rate_A	Cash_Flow_Per_Share	Per_Share_Net_profit_before_tax_Yuan_	Realized_Sales_Gross_Profit_Growth_Rate	Operating_Pro
0	-0.633296	1.539557	0.118477	-0.128462	-0.754347	0.053496	-0.748792	0.180520	
1	2.016795	0.135659	0.441752	-0.128462	-0.754347	0.164950	-0.373857	-0.082321	
2	-0.633296	0.177222	-0.141279	-0.128462	0.085538	0.112938	0.082794	-0.128307	
3	-0.633296	2.144527	-0.666470	-0.128462	-0.754347	-0.986745	-2.765267	-0.228174	
4	0.763746	0.301910	-0.050325	-0.128462	0.719997	0.578569	0.376013	-0.081416	

5 rows × 53 columns

Fig - 18 - Scaled Data

Model Building - Logistic Regression

Training Performance – Confusion Matrix



	Accuracy	Recall	Precision	F1
0	0.916397	0.393939	0.691489	0.501931

Fig – 19 Confusion Matrix | Performance Test data

- Accuracy (0.916397): This is the proportion of true results (both true positives and true negatives) among the total number of cases examined. It's a measure of how often the model is correct. An accuracy of 0.916397 means the model is correct about 91.64% of the time.
- Recall (0.393939): Also known as sensitivity, recall is the ratio of correctly predicted positive observations to all observations in actual class. It's a measure of a model's completeness. A recall of 0.393939 means that when the actual value is positive, the model correctly predicts a positive result about 39.39% of the time.
- Precision (0.691489): Precision is the ratio of correctly predicted positive observations to the total predicted positives. It's a measure of among all the predicted positive observations, how many of them are positive. A precision of 0.691489 means that when the model predicts a positive result, it's correct about 69.15% of the time.
- F1 Score (0.501931): The F1 Score is the weighted average of Precision and Recall. It tries to find the balance between precision and recall. An F1 score reaches its best value at 1 (perfect precision and recall) and worst at 0.

Confusion matrix insights:

- True Negatives (TN): These are the cases in which the model predicted ‘0’ and the actual class was also ‘0’. In your case, there are 1349 such instances, making up 87.43% of the total.
- False Positives (FP): These are the cases in which the model predicted ‘1’ but the actual class was ‘0’. There are 29 such instances in your data, making up 1.88% of the total.
- False Negatives (FN): These are the cases in which the model predicted ‘0’ but the actual class was ‘1’. There are 100 such instances in your data, making up 6.48% of the total.
- True Positives (TP): These are the cases in which the model predicted ‘1’ and the actual class was also ‘1’. There are 65 such instances in your data, making up 4.21% of the total.

General insights

In this confusion matrix, one can infer that the classifier performs well in identifying negative cases as it has a high true negative rate. However, there seems to be room for improvement in correctly identifying positive cases as indicated by the lower true positive rate compared to the true negative rate. False negatives are higher than false positives suggesting that it’s more likely for this model to miss out on actual positive cases than marking negative cases as positive falsely.

Test – Performance - Confusion Matrix

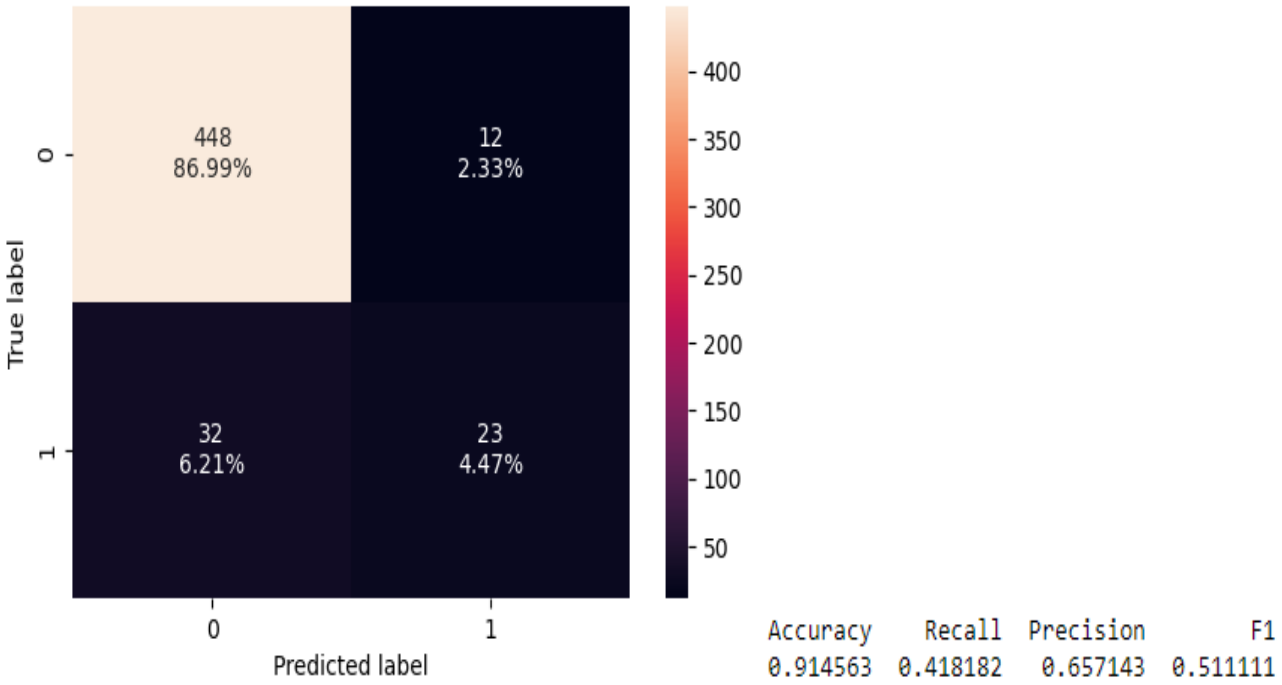


Fig – 20 Confusion Matrix | Performance Test data

Insights on Performance test

- Accuracy (0.914563): This is the proportion of true results (both true positives and true negatives) among the total number of cases examined. An accuracy of 0.914563 means that the model correctly predicted the outcome 91.4563% of the time across the test set.
- Recall (0.491812): This is the ability of the classifier to find all the positive samples. A recall of 0.491812 suggests that out of all actual positive cases in the test set, the model correctly identified 49.1812% as positive.
- Precision (0.657143): This is the ability of the classifier not to label a negative sample as positive. A precision of 0.657143 means that out of all cases predicted as positive by the model, 65.7143% were positive.
- F1 Score (0.511111): The F1 score is the harmonic mean of precision and recall. It tries to find the balance between precision and recall. An F1 score of 0.511111 indicates that there's room for improvement in achieving a better balance between precision and recall.

Insights on Confusion Matrix

- True Negatives (TN): These are the cases in which the model predicted ‘0’ and the actual class was also ‘0’. In your case, there are 448 such instances, making up 86.99% of the total.
- False Positives (FP): These are the cases in which the model predicted ‘1’ but the actual class was ‘0’. There are 12 such instances in your data, making up 2.33% of the total.
- False Negatives (FN): These are the cases in which the model predicted ‘0’ but the actual class was ‘1’. There are 32 such instances in your data, making up 6.21% of the total.
- True Positives (TP): These are the cases in which the model predicted ‘1’ and the actual class was also ‘1’. There are 23 such instances in your data, making up 4.47% of the total.

Random Forest – Model

Training Performance

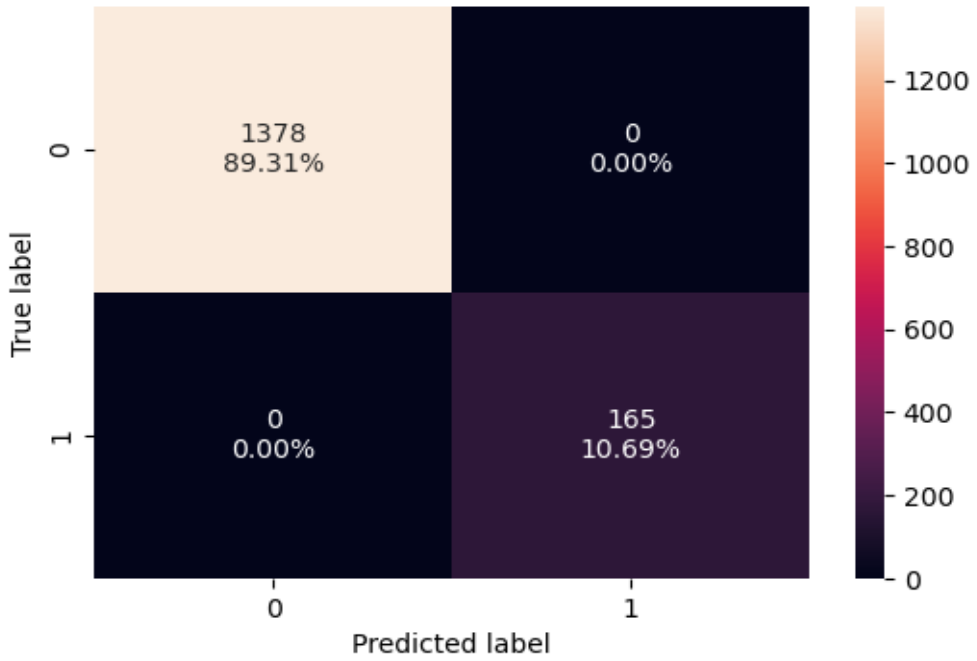


Fig – 21 Confusion Matrix

- True Negatives (TN): These are the cases in which the model predicted ‘0’ and the actual class was also ‘0’. In your case, there are 1378 such instances, making up 89.31% of the total.
- False Positives (FP): These are the cases in which the model predicted ‘1’ but the actual class was ‘0’. There are 0 such instances in your data, making up 0.00% of the total.
- False Negatives (FN): These are the cases in which the model predicted ‘0’ but the actual class was ‘1’. There are 0 such instances in your data, making up 0.00% of the total.
- True Positives (TP): These are the cases in which the model predicted ‘1’ and the actual class was also ‘1’. There are 165 such instances in your data, making up 10.69% of the total.

Random Forest Model Performance Classification.

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Fig – 22 Test performance results

- Accuracy (1.0): This is the proportion of true results (both true positives and true negatives) among the total number of cases examined. An accuracy of 1.0 means that the model correctly predicted the outcome 100% of the time across the test set.
- Recall (1.0): This is the ability of the classifier to find all the positive samples. A recall of 1.0 suggests that out of all actual positive cases in the test set, the model correctly identified 100% as positive.
- Precision (1.0): This is the ability of the classifier not to label a negative sample as positive. A precision of 1.0 means that out of all cases predicted as positive by the model, 100% were positive.
- F1 Score (1.0): The F1 score is the harmonic mean of precision and recall. It tries to find the balance between precision and recall. An F1 score of 1.0 indicates that the model has achieved a perfect balance between precision and recall.

These perfect scores across all metrics are highly desirable in machine learning, indicating that the Random Forest model has performed exceptionally well on the test set. However, it's worth noting that such perfect scores are rare in practice due to noise and other factors present in real-world data.

While these results are impressive, they might also raise some questions. For instance, if the model is performing with 100% effectiveness on the test set, it could be a sign of overfitting, where the model has become too complex and is fitting the noise in the training data, rather than the underlying pattern. This could lead to poor performance when the model is applied to new, unseen data.

Random Forest Model - Test Performance

Confusion matrix

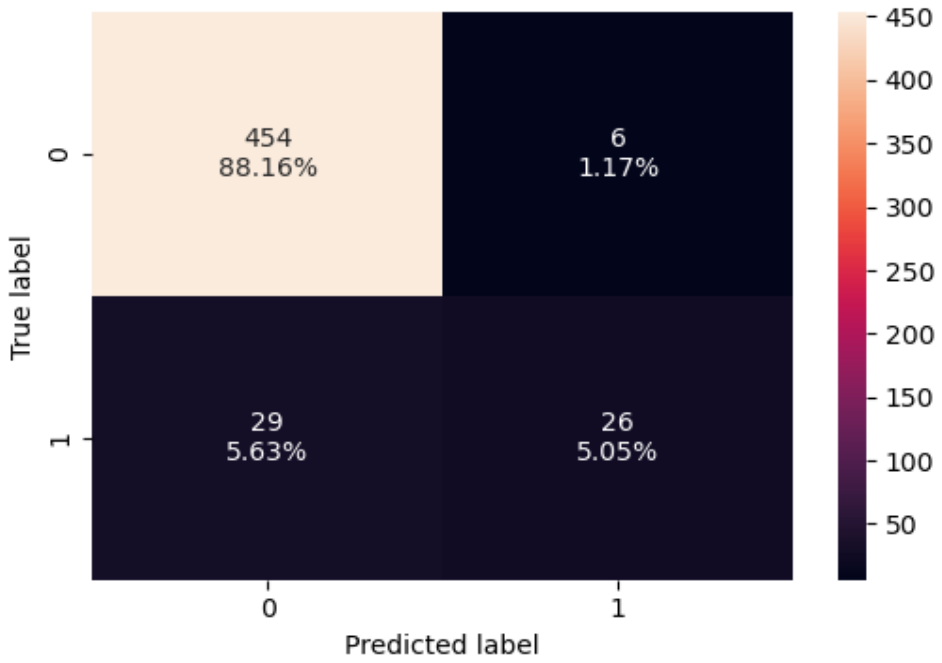


Fig – 23 Confusion Matrix – Random Forest Model - Test Performance

- True Negatives (TN): These are the cases in which the model predicted '0' and the actual class was also '0'. In your case, there are 454 such instances, making up 88.16% of the total.
- False Positives (FP): These are the cases in which the model predicted '1' but the actual class was '0'. There are 6 such instances in your data, making up 1.17% of the total.
- False Negatives (FN): These are the cases in which the model predicted '0' but the actual class was '1'. There are 29 such instances in your data, making up 5.63% of the total.
- True Positives (TP): These are the cases in which the model predicted '1' and the actual class was also '1'. There are 26 such instances in your data, making up 5.05% of the total.

Test model Performance.

	Accuracy	Recall	Precision	F1
0	0.932039	0.472727	0.8125	0.597701

Fig - 24 Model performance stats

- Accuracy (0.932039): This is the proportion of true results (both true positives and true negatives) among the total number of cases examined. An accuracy of 0.932039 means that the model correctly predicted the outcome 93.2039% of the time across the test set.
- Recall (0.472727): This is the ability of the classifier to find all the positive samples. A recall of 0.472727 suggests that out of all actual positive cases in the test set, the model correctly identified 47.2727% as positive.
- Precision (0.8125): This is the ability of the classifier not to label a negative sample as positive. A precision of 0.8125 means that out of all cases predicted as positive by the model, 81.25% were positive.
- F1 Score (0.597701): The F1 score is the harmonic mean of precision and recall. It tries to find the balance between precision and recall. An F1 score of 0.597701 indicates that there's room for improvement in achieving a better balance between precision and recall.

From these metrics, we can infer that while the model has high accuracy, it seems to struggle with recall, which could indicate a high number of false negatives. In other words, the model might be predicting many actual positive cases as negative. The precision is relatively better, indicating that when the model predicts a positive case, it is correct about 81.25% of the time. However, the F1 score, which balances precision and recall, is just above 0.5, suggesting that the model could be improved in terms of its ability to correctly identify positive cases (increasing recall) without incorrectly labelling negative cases as positive (maintaining or increasing precision).

Model Performance Improvement

Calculate Variance Inflation Factor

Variance Inflation Factors:

	Variable	VIF
0	Operating_Expense_Rate	1.259611
1	Research_and_development_expense_rate	1.099306
2	Cash_flow_rate	12.259184
3	Interest_bearing_debt_interest_rate	1.032597
4	Tax_rate_A	1.248346
5	Cash_Flow_Per_Share	4.564430
6	Per_Share_Net_profit_before_tax_Yuan_	8.756612
7	Realized_Sales_Gross_Profit_Growth_Rate	1.058517
8	Operating_Profit_Growth_Rate	1.152581
9	Continuous_Net_Profit_Growth_Rate	1.467948
10	Total_Asset_Growth_Rate	1.174794
11	Net_Value_Growth_Rate	1.044367
12	Total_Asset_Return_Growth_Rate_Ratio	1.134845
13	Cash_Reinvestment_perc	7.340538
14	Current_Ratio	4.945713
15	Quick_Ratio	1.063685
16	Interest_Expense_Ratio	1.033551
17	Total_debt_to_Total_net_worth	3.776391
18	Long_term_fund_suitability_ratio_A	1.839945
19	Net_profit_before_tax_to_Paid_in_capital	8.637685
20	Total_Asset_Turnover	5.467530
21	Accounts_Receivable_Turnover	1.064519
22	Average_Collection_Days	1.060724
23	Inventory_Turnover_Rate_times	1.100171
24	Fixed_Assets_Turnover_Frequency	1.223623
25	Net_Worth_Turnover_Rate_times	3.945259
26	Operating_profit_per_person	1.568575
27	Allocation_rate_per_person	1.198618
28	Quick_Assets_to_Total_Assets	2.397607
23	Inventory_Turnover_Rate_times	1.100171
24	Fixed_Assets_Turnover_Frequency	1.223623
25	Net_Worth_Turnover_Rate_times	3.945259
26	Operating_profit_per_person	1.568575
27	Allocation_rate_per_person	1.198618
28	Quick_Assets_to_Total_Assets	2.397607
29	Cash_to_Total_Assets	2.183010
30	Quick_Assets_to_Current_Liability	1.009579
31	Cash_to_Current_Liability	1.079209
32	Operating_Funds_to_Liability	12.536226
33	Inventory_to_Working_Capital	1.459350
34	Inventory_to_Current_Liability	1.124100
35	Long_term_Liability_to_Current_Assets	1.102010
36	Retained_Earnings_to_Total_Assets	3.365775
37	Total_income_to_Total_expense	1.676735
38	Total_expense_to_Assets	3.366230
39	Current_Asset_Turnover_Rate	1.416203
40	Quick_Asset_Turnover_Rate	1.377544
41	Cash_Turnover_Rate	1.107230
42	Fixed_Assets_to_Assets	1.815190
43	Cash_Flow_to_Total_Assets	3.309496
44	Cash_Flow_to_Liability	2.813638
45	CFO_to_Assets	10.987676
46	Cash_Flow_to_Equity	1.425404
47	Current_Liability_to_Current_Assets	1.464422
48	Total_assets_to_GNP_price	1.041114
49	No_credit_Interval	1.032531
50	Degree_of_Financial_Leverage_DFL	1.015012
51	Interest_Coverage_Ratio_Interest_expense_to_EBIT	1.018535
52	Equity_to_Liability	4.779776

Fig - 25 Variance Inflation Factor

Identify optimal threshold for Logistic Regression using ROC curve.

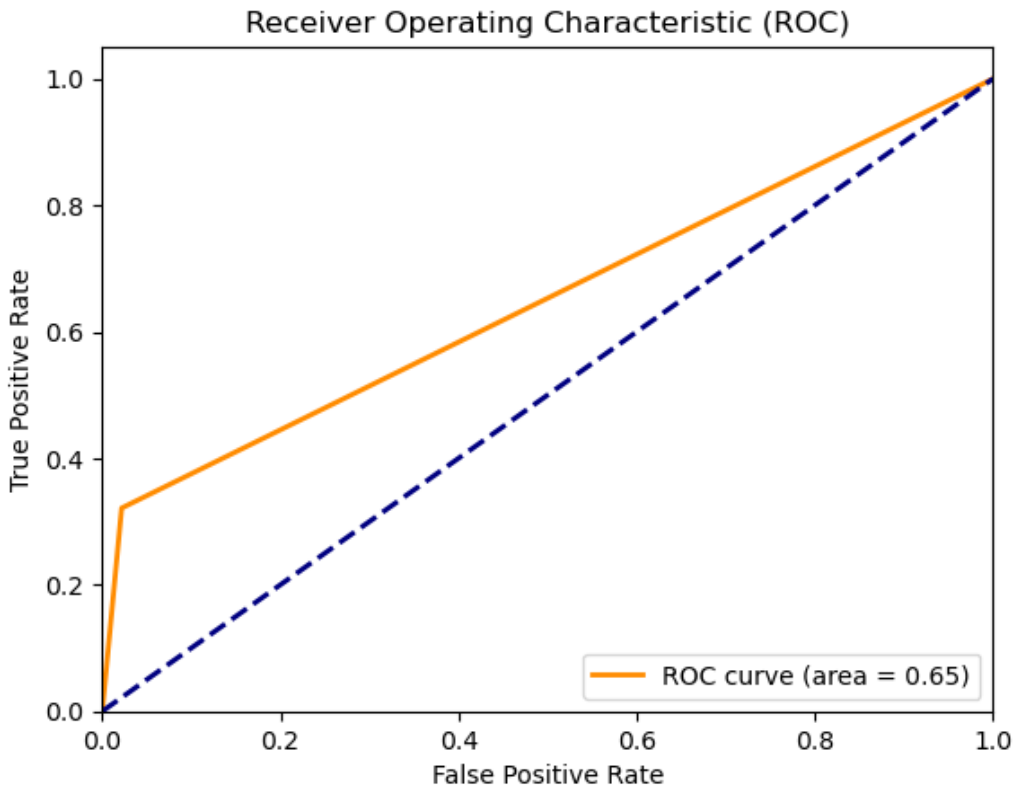


Fig - 26 ROC Analysis

True Positive Rate (TPR): This measures how many actual positives are correctly identified by the model. It's also known as sensitivity or recall.

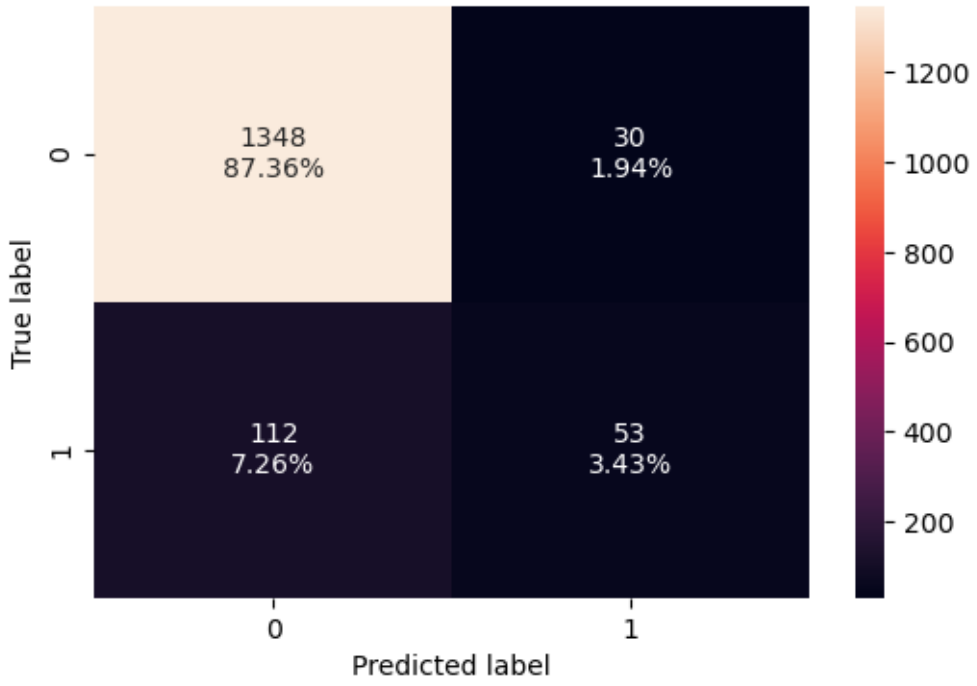
False Positive Rate (FPR): This measures how many actual negatives are incorrectly identified as positives by the model.

Insights

The ROC curve in your image shows the performance of a binary classifier. The curve is plotted with TPR against the FPR at various threshold settings. The area under the ROC curve (AUC) is a measure of how well a parameter can distinguish between two diagnostic groups. In your case, the AUC is 0.65. This suggests that there's a 65% chance that the model will be able to distinguish between positive class and negative class. Generally, an AUC of 0.5 suggests no discrimination (i.e., ability to diagnose patients correctly), 0.7 to 0.8 is considered acceptable, 0.8 to 0.9 is considered excellent, and more than 0.9 is considered outstanding. The diagonal dashed line represents the ROC curve of a purely random classifier; a good classifier stays as far away from that line as possible (toward the top-left corner).From the ROC curve, it looks like your model does a reasonable job of distinguishing the positive from the negative class. However, there's still room for improvement as the AUC is not close to 1.

Model Performance check post VIF & ROC

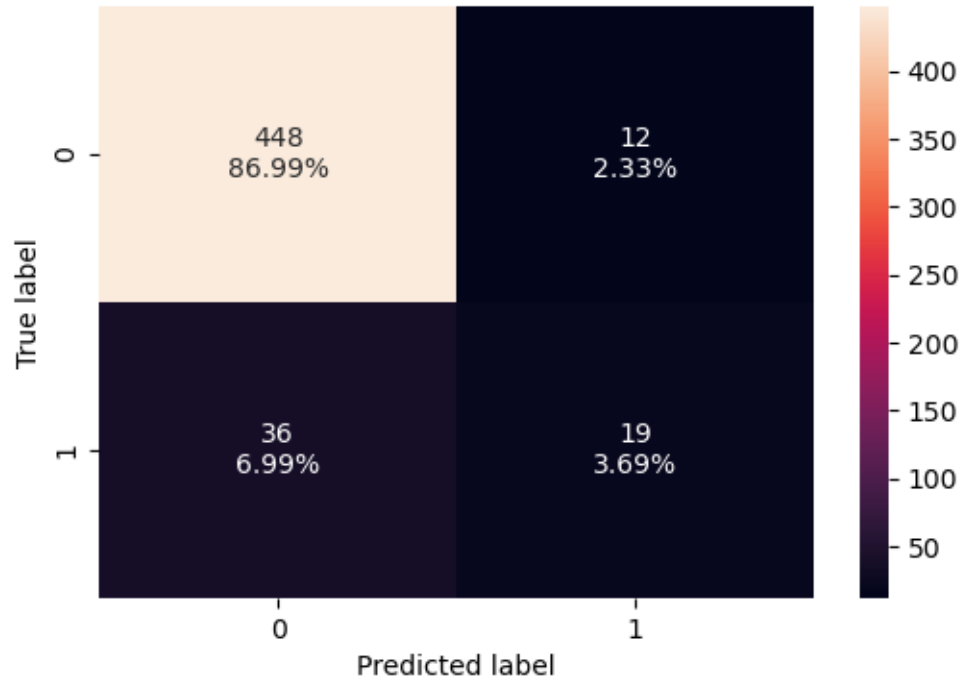
Training Set



	Accuracy	Recall	Precision	F1
0	0.907971	0.321212	0.638554	0.427419

Fig – 27 Confusion Matrix | Model performance stats

Test Set



	Accuracy	Recall	Precision	F1
0	0.906796	0.345455	0.612903	0.44186

Fig – 28 Confusion Matrix Test set | Model performance stats

Here are some insights and conclusions we can draw:

1. **Model Performance**: The first model has a higher True Positive rate (86.99%) compared to the second model (60.78%). This suggests that the first model is better at correctly identifying positive cases.
2. **False Positives**: The second model has a significantly higher False Positive rate (28.54%) compared to the first model (2.33%). This means the second model is more likely to incorrectly identify a negative case as positive.
3. **False Negatives**: Both models have similar False Negative rates, with the first model at 6.02% and the second model at 6.60%. This means both models have a similar likelihood of incorrectly identifying a positive case as negative.
4. **True Negatives**: The first model has a slightly higher True Negative rate (4.66%) compared to the second model (4.08%). This suggests that the first model is slightly better at correctly identifying negative cases.

PART B

Executive Overview

The dataset contains data on the weekly share prices of five Indian companies over a duration of eight years. This data provides an opportunity to explore the past performance of these individual stocks and understand the wider trends in the market.

Objective

The objective of this analysis is to conduct Market Risk Analysis on a portfolio of Indian stocks using Python. It uses historical stock price data to understand market volatility and riskiness. Using statistical measures like mean and standard deviation, investors gain a deeper understanding of individual stocks' performance and portfolio variability.

Through this analysis, investors can aim to achieve the following objectives:

1. Risk Assessment: Analyze historical volatility of individual stocks and the overall portfolio.
2. Portfolio Optimization: Use Market Risk Analysis insights to enhance risk-adjusted returns.
3. Performance Evaluation: Assess portfolio management strategies' effectiveness in mitigating market risk.
4. Portfolio Performance Monitoring: Monitor portfolio performance over time and adjust as market conditions and risk preferences change.

Data dictionary: The dataset contains weekly stock price data for 5 Indian stocks over an 8-year period. The dataset enables us to analyse the historical performance of individual stocks and the overall market dynamics.

Sample of the Dataset

	Date	Dish TV	Infosys	Hindustan Unilever	Vodafone Idea	Cipla
0	28-03-2016	86	608	867	67	514
1	04-04-2016	86	607	863	65	519
2	11-04-2016	85	583	853	66	506
3	18-04-2016	87	625	900	69	515
4	25-04-2016	89	606	880	71	532

Fig – 1

Data Shape

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 6 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Date                418 non-null    object
 1   Dish TV             418 non-null    int64
 2   Infosys             418 non-null    int64
 3   Hindustan Unilever  418 non-null    int64
 4   Vodafone Idea       418 non-null    int64
 5   Cipla               418 non-null    int64
dtypes: int64(5), object(1)
memory usage: 19.7+ KB
```

Fig - 2

Statistical Summary

	Dish TV	Infosys	Hindustan Unilever	Vodafone Idea	Cipla
count	418.000000	418.000000	418.000000	418.000000	418.000000
mean	38.648325	1007.210526	1906.344498	23.234450	756.614833
std	31.944620	455.089501	597.800173	20.264854	252.969619
min	4.000000	445.000000	788.000000	3.000000	370.000000
25%	14.000000	591.250000	1368.500000	9.000000	556.000000
50%	19.500000	777.500000	2083.000000	12.000000	637.000000
75%	73.000000	1454.000000	2419.000000	43.000000	946.000000
max	108.000000	1939.000000	2798.000000	71.000000	1493.000000

Fig – 3

- Count: All stocks have 418 observations, indicating the dataset is complete with weekly data for each stock over the 8-year period.
- Mean: The average weekly prices differ significantly across the stocks, reflecting their varied market positions and valuations.
- Standard Deviation (std): This measures the volatility or spread of the stock prices. Higher std indicates greater price variability.
- Min and Max: These values show the lowest and highest weekly prices over the period, respectively.
- Percentiles (25%, 50%, 75%): These provide a sense of the price distribution. The 50th percentile (median) is particularly useful for understanding the central tendency of the data.

Insights

1. Dish TV:
    - Mean Price: 38.65
    - Standard Deviation: 31.94
    - Min and Max: 4.00 and 108.00
    - Percentiles: 14.00 (25%), 19.50 (50%), 73.00 (75%)
    - Insights:

- The mean price of Dish TV is relatively low, suggesting it might be a lower-valued stock compared to others.
- The high standard deviation relative to the mean indicates considerable volatility in Dish TV's stock price.
- The large gap between the 25th and 75th percentiles (14 to 73) shows that the stock price has significant variability.

2. Infosys:
    - Mean Price: 1007.21
    - Standard Deviation: 455.09
    - Min and Max: 445.00 and 1939.00
    - Percentiles: 591.25 (25%), 777.50 (50%), 1454.00 (75%)
    - Insights:
        - Infosys has a higher mean price, indicating a relatively high valuation in the market.
        - The substantial standard deviation reflects high price volatility.
        - The median (777.50) is lower than the mean, suggesting a right-skewed distribution with some high price outliers.
3. Hindustan Unilever:
    - Mean Price: 1906.34
    - Standard Deviation: 597.80
    - Min and Max: 788.00 and 2798.00
    - Percentiles: 1368.50 (25%), 2083.00 (50%), 2419.00 (75%)
    - Insights:
        - Hindustan Unilever has the highest mean price among the five stocks, indicating a strong market position.
        - A high standard deviation suggests significant price volatility.
        - The median (2083.00) is higher than the mean, suggesting a left-skewed distribution with more frequent lower prices.

4. Vodafone Idea:
    - Mean Price: 23.23
    - Standard Deviation: 20.26
    - Min and Max: 3.00 and 71.00
    - Percentiles: 9.00 (25%), 12.00 (50%), 43.00 (75%)
    - Insights:
        - Vodafone Idea has the lowest mean price, indicating it is a lower-valued stock.
        - A high standard deviation relative to the mean shows high volatility.
        - The small gap between the median and the 75th percentile suggests more stable prices within the lower range, with some higher price spikes.
5. Cipla:
    - Mean Price: 756.61
    - Standard Deviation: 252.97
    - Min and Max: 370.00 and 1493.00
    - Percentiles: 556.00 (25%), 637.00 (50%), 946.00 (75%)
    - Insights:
        - Cipla's mean price is moderate compared to other stocks, reflecting its mid-to-high market valuation.
        - The standard deviation indicates moderate volatility.
        - The median (637.00) is lower than the mean, suggesting a right-skewed distribution with some high price outliers.

Summary Insights:

- Price Variability: All stocks exhibit significant variability, as indicated by their standard deviations. This is typical in the stock market, reflecting the underlying volatility.
- Market Valuation: Hindustan Unilever and Infosys are high-valued stocks with mean prices over 1000, indicating strong market positions. Vodafone Idea and Dish TV have much lower mean prices, suggesting lower market valuations.

- Distribution Skewness: Infosys and Cipla show right-skewed distributions, meaning there are occasional high prices. Hindustan Unilever shows a left-skewed distribution with frequent lower prices but some very high prices.
- Volatility: Hindustan Unilever and Infosys have the highest standard deviations, indicating they are more volatile compared to Cipla, Dish TV, and Vodafone Idea.

Dataset Processing

Null value identification –

No null values found in the data set.

```
Date                0
Dish TV             0
Infosys             0
Hindustan Unilever  0
Vodafone Idea       0
Cipla               0
dtype: int64
```

Fig – 4

Company wise Share price vs Time

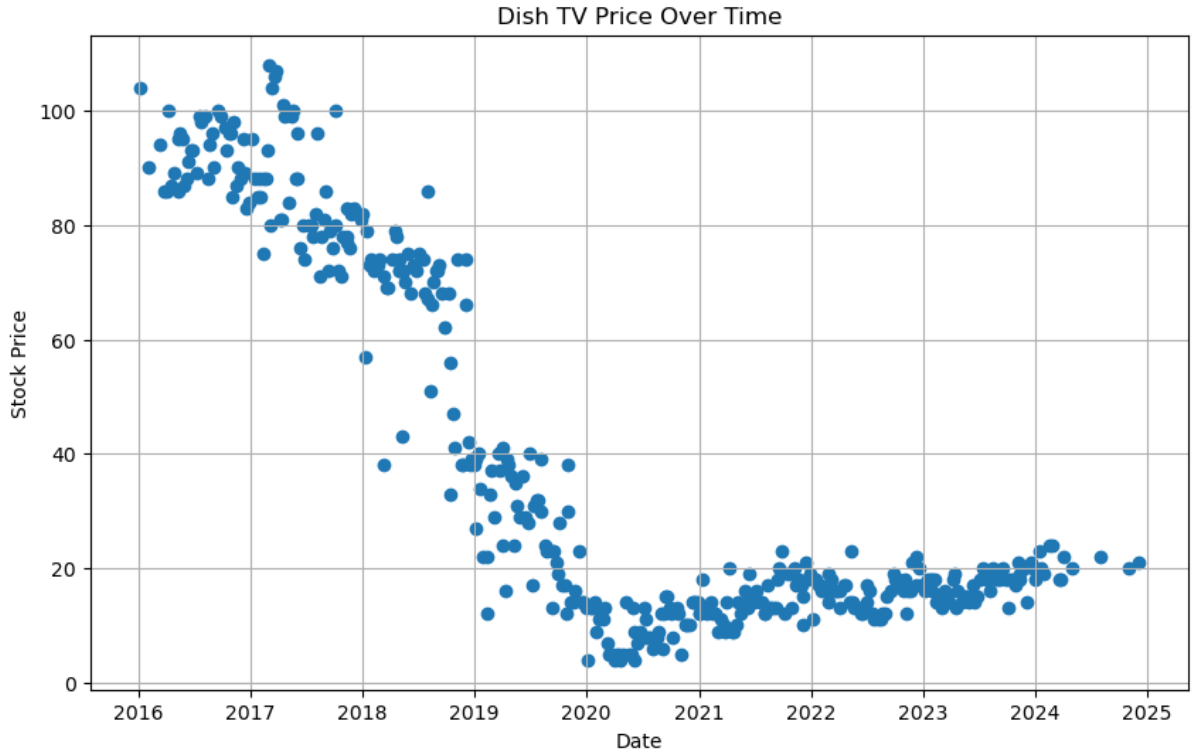


Fig - 5 Dish TV share price over time.

The graph is a scatter plot of Dish TV's stock price over time. The x-axis represents the time from 2016 to 2025, and the y-axis represents the stock price ranging from 0 to above 100.

Here are some insights:

1. Downward Trend: The most noticeable trend in the graph is the significant decline in Dish TV's stock price over time. The stock price was fluctuating around 80 to just over 100 in 2016, but it has decreased to around or below 20 in recent years. This suggests that the company's market value has been decreasing over this period.
2. Volatility: The spread of the data points at each period indicates the volatility of the stock price. There seems to be a high volatility in the earlier years (2016-2018) where the stock price fluctuated between a wide range of values. In contrast, the volatility seems to have decreased in the later years (2023-2025) where the stock prices are mostly clustering at the lower end of the scale.
3. Investment Considerations: The sharp decrease in stock price suggests that investors may have lost confidence in Dish TV's financial health or prospects. The concentration

of lower prices towards recent years could be indicative of ongoing challenges within the company or industry.

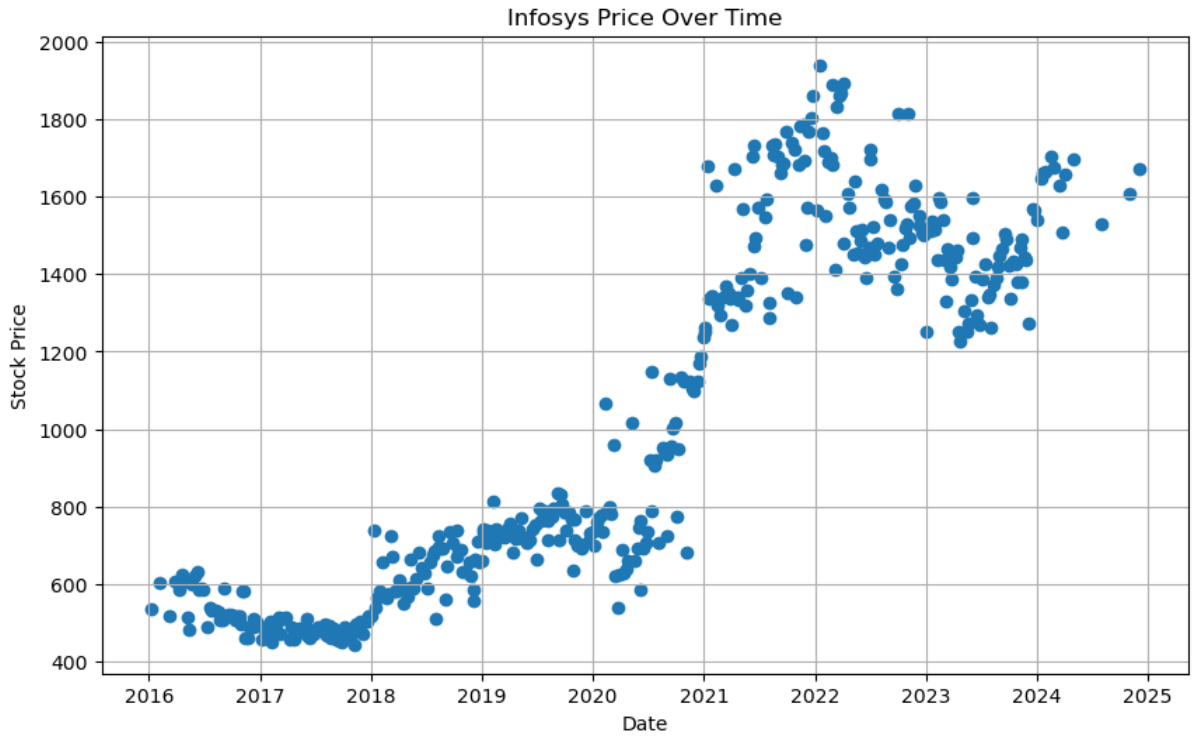


Fig – 6 Scatterplot for Infosys share price over time

Here are some key observations and insights based on the plot:

1. Stability (2016 - Early 2020): From around 2016 until early 2020, the stock price remained relatively stable, fluctuating between roughly 900 and 1200. This suggests a period of stability in the company's performance and market perception.
2. Growth (Early 2020 - 2025): Starting in early 2020, there's a noticeable upward trend where the stock price increases significantly over time. This rise continues with some fluctuations but maintains an overall upward trajectory through to the end of the graph in 2025. This could be indicative of a period of growth for Infosys, possibly due to various factors such as company performance improvements, market conditions favoring technology stocks, or strategic business decisions made by Infosys during that period.
3. Fluctuations: Throughout the entire period, there are fluctuations in the stock price. These fluctuations could be due to a variety of factors including market volatility, economic indicators, company-specific news, and financial results.

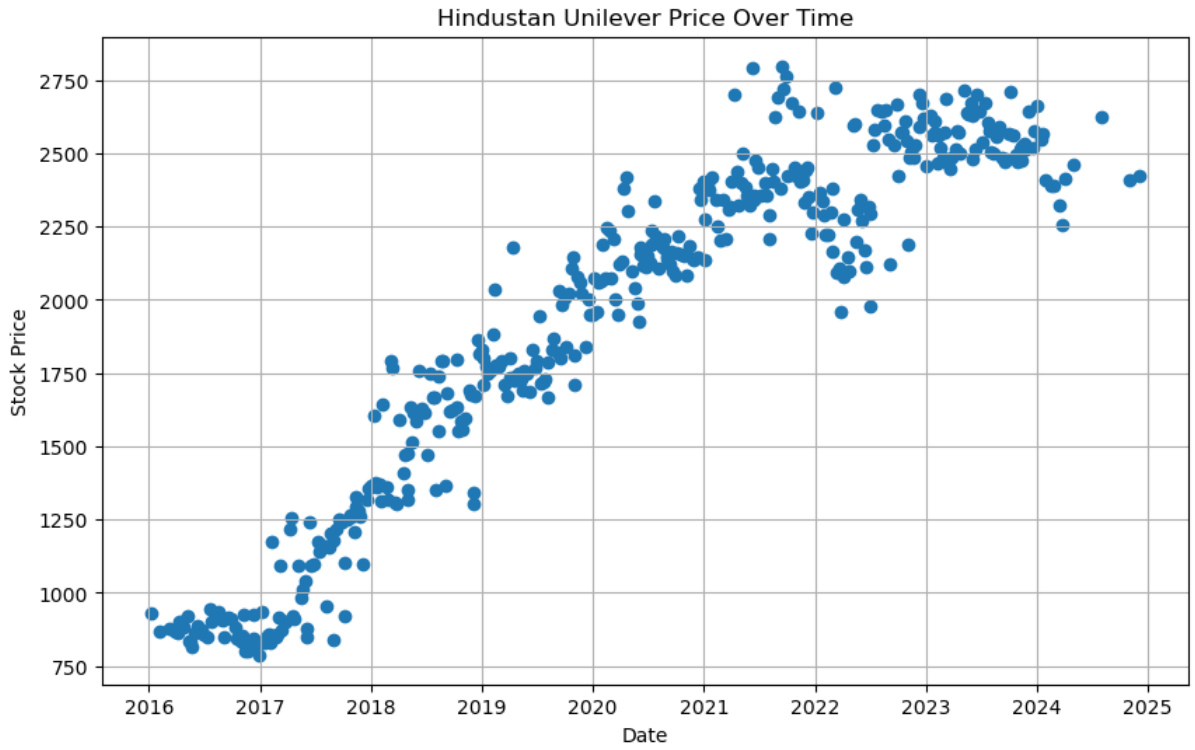


Fig - 7 Scatterplot for Hindustan Unilever share price over time

1. Steady Increase (2016 - 2024): The stock price starts just above 1000 INR in 2016 and shows a steady increase throughout the years, reaching up to around 2750 INR by 2024. This indicates that Hindustan Unilever's market performance has been positive over these years.
2. Fluctuations: Throughout the entire period, there are fluctuations in the stock price. These fluctuations could be due to a variety of factors including market volatility, economic indicators, company-specific news, and financial results.
3. Overall Upward Trend: Despite these fluctuations, the overall trend of the stock price is upward. This suggests that the company has been growing steadily over this period.

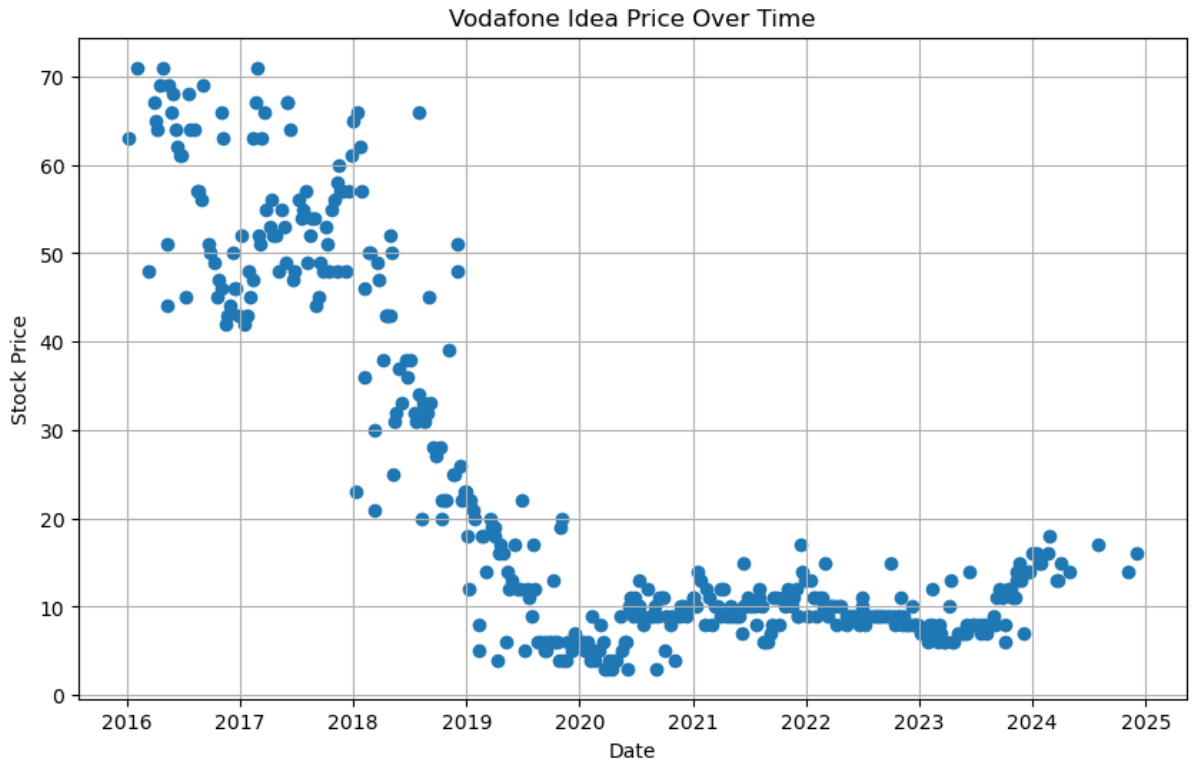


Fig – 8 Scatterplot for Vodafone Idea share price over time.

1. High Share Price (2016 - Early 2018): The plot shows a high concentration of data points above 50 on the vertical axis between 2016 and early 2018, suggesting that during this period, Vodafone Idea's share price was relatively high.
2. Sharp Decline (Early 2018 - Mid 2020): Starting from early 2018, there is a noticeable downward trend where data points begin to drop sharply below 20 and continue to decline until mid-2020. This could be indicative of a period of significant challenges for Vodafone Idea, possibly due to factors such as market conditions, company performance issues, or shifts in investor confidence.
3. Stabilization (Mid 2020 - 2025): From mid-2020 onwards, the share price stabilizes. The data points are densely packed around a narrow band between approximately 5 and 15 on the vertical axis up until 2025. This indicates that after a period of decline, Vodafone Idea's share price has leveled out at a lower value compared to previous years.

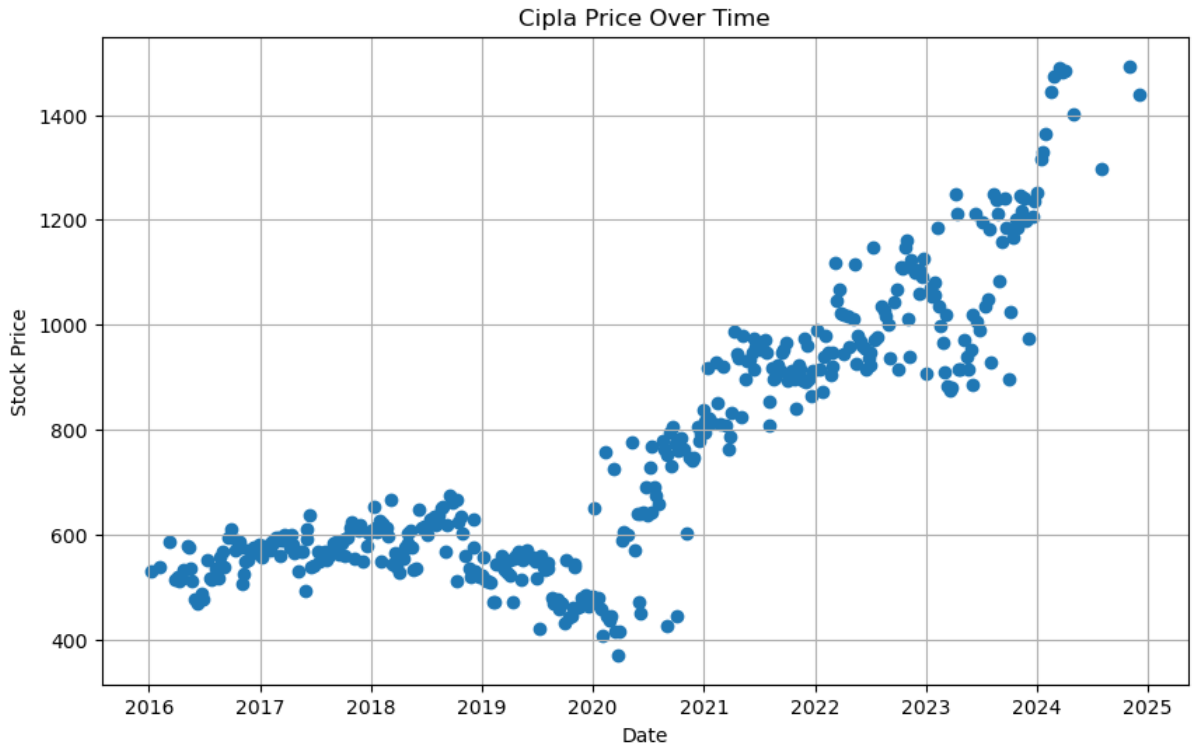


Fig – 9 Scatterplot for Cipla share price over time

1. Stable Share Price (2016 - Late 2019): For a significant period between 2016 and late 2019, Cipla's share price remained relatively stable, fluctuating around the lower end of the price range on this graph (approximately between 600 and 800). This suggests that during this period, Cipla's market performance was relatively stable.
2. Sharp Increase (Early to Mid-2020): Starting around early to mid-2020, there is a noticeable upward trend in share prices. This trend continues sharply upwards through to what appears to be mid-2024. This could be indicative of a period of significant growth for Cipla, possibly due to factors such as market conditions, company performance improvements, or shifts in investor confidence.
3. Increased Volatility (Mid-2020 Onwards): The concentration of points becomes denser as time progresses towards later dates and higher prices, indicating increased volatility or more frequent trading as prices rise.

Average Returns

```
Vodafone Idea        -0.003932
Dish TV              -0.003751
Infosys               0.002180
Hindustan Unilever    0.002294
Cipla                 0.002538
dtype: float64
```

Fig – 10 Average Returns from stocks under analysis

1. Vodafone Idea: The return is -0.003932. This negative return indicates that the stock's value has decreased over the period under consideration.

2. Dish TV: The return is -0.003751. Like Vodafone Idea, this negative return suggests a decrease in the stock's value.

3. Infosys: The return is 0.002180. This positive return indicates an increase in the stock's value.

4. Hindustan Unileer: The return is 0.002294. This is another stock that has increased in value, as indicated by the positive return.

5. Cipla: The return is 0.002538. This stock has the highest return among the five, suggesting it has performed the best over the period.

Volatility Of the Stocks

```
Hindustan Unilever    0.028845
Infosys               0.036102
Cipla                 0.036759
Dish TV               0.091333
Vodafone Idea         0.113747
dtype: float64
```

Fig – 11 Volatility across data

1. Hindustan Unilever: The standard deviation is 0.028845. This relatively low value suggests that the returns of Hindustan Unilever have been stable, with less fluctuation in price.
2. Infosys: The standard deviation is 0.036102. This indicates a slightly higher volatility compared to Hindustan Unilever, meaning the returns for Infosys have been more variable.
3. Cipla: The standard deviation is 0.036759. This is like Infosys, suggesting a comparable level of volatility.
4. Dish TV: The standard deviation is 0.091333. This significantly higher value indicates that Dish TV's returns have been quite volatile, with a larger fluctuation in price.
5. Vodafone Idea: The standard deviation is 0.113747. This is the highest among the five stocks, indicating that Vodafone Idea has had the most volatile returns.

In market analysis, standard deviation is a measure of risk. A higher standard deviation means the price of the stock is moving significantly, and thus, the stock carries more risk. Conversely, a lower standard deviation means the price movements are less volatile, and the stock is less risky.

You might note that while Vodafone Idea and Dish TV have the highest volatility, they may also present the greatest potential for high returns. However, they also carry a higher risk. On the other hand, Hindustan Unilever, with the lowest volatility, presents the least risk but may also offer lower potential returns.

Visualizing Returns and Volatility

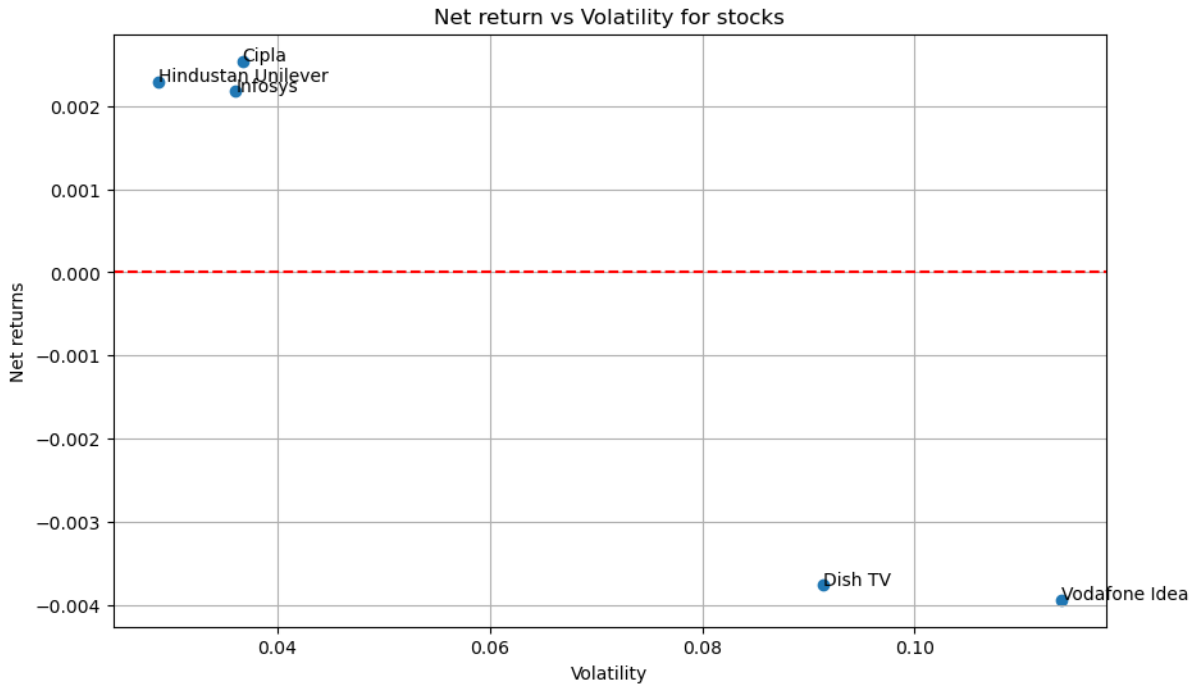


Fig – 12 Plot of Net returns Vs Volatility

1. Cipla: This stock has moderate volatility and positive net returns. This suggests that while the stock's price does fluctuate, it has generally provided positive returns.
2. Hindustan Unilever: This stock has relatively low volatility and positive net returns, indicating that it has been a stable investment with consistent positive returns.
3. Dish TV: This stock is an outlier with negative net returns and moderate volatility. This suggests that the stock's price has been fluctuating, and it has generally provided negative returns.
4. Vodafone Idea: This stock is another outlier with positive net returns but high volatility. This suggests that while the stock has provided positive returns, it has been a risky investment due to its high price fluctuations.

- Observations:
    - Positive Returns: Stocks with positive returns vary in their levels of volatility. Interestingly, some stocks manage to yield positive returns despite their volatility.
    - Negative Returns: On the other hand, stocks with negative returns often exhibit high volatility, indicating a higher level of risk. Investors should exercise caution when considering these stocks.

- Inferences:
    - The graph highlights the trade-off between risk (as measured by volatility) and potential returns. Some stocks can yield positive returns despite their volatility, while others fail to do so.
    - It's important for investors to carefully assess their risk tolerance and consider diversification strategies to manage risk effectively.
    - Investment decisions are heavily influenced by individual preferences, investment goals, and risk appetite. Diversification across various stocks can help manage risk and potentially enhance returns.